

Plant Disease Classification

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Summary: Performance metrics across multiple datasets using various deep learning architectures and transformation pipelines.

1. Datasets Overview

A summary of the datasets used for training, testing and validation.

Dataset Name	Size (Records)	Available Classes	Trained and Tested the Model On
New Plant Disease Dataset (Augmented)	87,000	38	10 classes
Plant-Doc	2,572	28	10 classes, 27 classes
Plant-Wild v1	18,542	89	10 classes
Rice Disease Dataset	45,400	19	19 classes

2. Models Overview

Efficient-Net => EfficientNet is a family of convolutional neural networks (CNNs) for computer vision published by researchers at Google AI in 2019. Its key innovation is **compound scaling**, which uniformly scales all dimensions of depth, width, and resolution using a single parameter.

Convolutional-Next => Constructed entirely from standard ConvNet modules, **ConvNeXts** compete favorably with Transformers in terms of accuracy and scalability, achieving **87.8% ImageNet top-1 accuracy** and outperforming **Swin Transformers** on **COCO detection** and **ADE20K segmentation**, while maintaining the simplicity and efficiency of standard ConvNets.

YOLO-26 => YOLO26 is the latest evolution in the YOLO series of real-time object detectors, engineered from the ground up for **edge and low-power devices**. It introduces a streamlined design that removes unnecessary complexity while integrating targeted innovations to deliver faster, lighter, and more accessible deployment.

Inception V3 => Inception is a family of convolutional neural network (CNN) for computer vision, introduced by researchers at **Google** in 2014 as **GoogLeNet** (later renamed **Inception v1**). The series was historically important as an early CNN that separates the stem (data ingest), body (data processing), and head (prediction), an architectural design that persists in all modern CNN.

VGG 16 => The **VGGNets** are a series of convolutional neural networks (CNNs) developed by the Visual Geometry Group (VGG) at the University of Oxford. The **VGG** family includes various configurations with different depths, denoted by the letter “**VGG**” followed by the number of weight layers. The most common ones are **VGG-16** and **VGG-19**.

3. Best Performing Models

Conv-Next emerged out to be the best performing model for classifying both Potato+Tomato subclasses in the **Plant_Doc+ Plant_Wild dataset** as well as the diseases under **Rice Disease Dataset**.

Dataset	Transformations Applied	Model Used	Optimizer	Epochs	Accuracy	Precision	F1-Score
Plant_Doc + Plant_Wild [10 Classes (Potato+Tomato)]	Resized to 224, Rotation (0-180°), Histogram Equalization, YOLO v11 cropping, RandomHorizontalFlip	Conv-Next Base	AdamW	20	60.21%	0.58	0.56
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation (0-180°), RandomHorizontalFlip	Conv-Next Base	AdamW	20	94.19%	0.95	0.94

4. Comprehensive Results Table

For Potato + Tomato Subclasses [10 Classes]:

Testing Configuration

Evaluated on: *Potato (Early blight, healthy, Late blight)* and *Tomato (Bacterial spot, Early blight, healthy, Late blight, Leaf mold, Septoria leaf spot, Yellow Leaf Curl Virus)*.

Dataset	Transformations Applied	Model Used	Optimizer	Epochs	Testing Accuracy
New Plant Diseases + Plant_Doc + Plant_Wild [10 Classes]	Resized to 224, YOLO v11 cropping, RandomHorizontalFlip	Efficient Net B4	AdamW	20	57.72%
New Plant Diseases + Plant_Doc + Plant_Wild [10 Classes]	Resized to 380, YOLO v11 cropping, RandomHorizontalFlip	Efficient Net B4	AdamW	20	54.51%
Plant_Doc + Plant_Wild [10 Classes]	Resized to 224, YOLO v11 cropping, RandomHorizontalFlip	Efficient Net B4	AdamW*	40	55.58%
Plant_Doc + Plant_Wild [10 Classes]	Resized to 224, YOLO v11 cropping, RandomHorizontalFlip	Efficient Net B7	AdamW*	40	52.85%
Plant_Doc + Plant_Wild [10 Classes]	Resized to 224, YOLO v11 cropping, RandomHorizontalFlip	Conv-Next Large	AdamW	10	56.77%
Plant_Doc + Plant_Wild [10 Classes]	Resized to 299, Rotation, Hist. Eq, YOLO v11, RandomHorizontalFlip	Conv-Next Base	AdamW	20	59.86%
Plant_Doc + Plant_Wild [10 Classes]	Resized to 224, Rotation, Hist. Eq, YOLO v11, RandomHorizontalFlip	Conv-Next Small	AdamW	20	55.34%
Plant_Doc + Plant_Wild [Potato Subclasses Only]	Resized to 224, YOLO v11 cropping, RandomHorizontalFlip	Efficient Net B4	AdamW	20	69.29%
Plant_Doc + Plant_Wild [Potato Subclasses Only]	Resized to 224, Rotation, YOLO v11, RandomHorizontalFlip	Conv-Next Base	AdamW	20	73.62%
Plant_Doc + Plant_Wild [Tomato Subclasses Only]	Resized to 224, YOLO v11 cropping, RandomHorizontalFlip	Efficient Net B7	AdamW	20	59.86%

Dataset	Transformations Applied	Model Used	Optimizer	Epochs	Testing Accuracy
Plant_Doc + Plant_Wild [Tomato Subclasses Only]	Resized to 224, YOLO v11 cropping, RandomHorizontalFlip	Conv- Next Base	AdamW	20	66.50%
Plant_Doc + Plant_Wild [10 Classes]	Resized to 224, YOLO v11, Rotation, Flip, Hist. Eq	YOLO26- L	AdamW	100	51.7%

For Rice Disease Dataset

Testing Configuration

Evaluated comprehensively across all **19 classes**.

Dataset	Transformations Applied	Model Used	Optimizer	Epochs	Testing Accuracy
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, WeightedRandomSampling, Flip	Conv- Next Base	AdamW	10	93.98%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, Hist. Eq, RandomHorizontalFlip	Conv- Next Base	AdamW	20	91.40%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, WeightedRandomSampling, Flip	Conv- Next Base	AdamW	20	91.40%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, RandomHorizontalFlip	Efficient Net B4	AdamW	10-20	89.68%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, Center Cropping, RandomHorizontalFlip	Conv- Next Base	AdamW	20	88.39%

Dataset	Transformations Applied	Model Used	Optimizer	Epochs	Testing Accuracy
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, RandomHorizontalFlip	Conv-Next Large	AdamW	12/15	87.74%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, RandomHorizontalFlip, WeightedRandomSampling	Conv-Next Base	AdamW	20	91.40%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, RandomHorizontalFlip, WeightedRandomSampling	Conv-Next Base	AdamW	10	93.98%
Rice Disease Dataset [19 Classes]	Resized to 299, Rotation, RandomHorizontalFlip	Inception V3	AdamW	30	93.12%
Rice Disease Dataset [19 Classes]	Resized to 299, Rotation, Flip, WeightedRandomSampling	Inception V3	AdamW	30	92.04%
Rice Disease Dataset [19 Classes]	Resized to 299, Rotation, Flip, WeightedRandomSampling	VGG 16	AdamW	30	90.11%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, Flip, WeightedRandomSampling	VGG 16	AdamW	30	89.68%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, RandomHorizontalFlip	VGG 16	AdamW	30	83.66%
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, Flip, Brown Spot replacement source	Conv-Next-Base	AdamW	20	86.67%

Dataset	Transformations Applied	Model Used	Optimizer	Epochs	Testing Accuracy
Rice Disease Dataset [19 Classes]	Resized to 224, Rotation, RandomHorizontalFlip	YOLO26-L	AdamW	100	88.2%

For Plant_Doc Test Dataset[27 Classes]

Testing Configuration

Includes: Apple (Scab, leaf, rust), Bell_pepper (leaf, spot), Blueberry, Cherry, Corn (Gray leaf spot, leaf blight, rust), Grape (leaf, black rot), Peach, Potato (early/late blight), Raspberry, Soyabean, Squash (Powdery mildew), Strawberry, and Tomato (7 variants).

Dataset	Transformations Applied	Model Used	Optimizer	Epochs	Testing Accuracy
Plant_Doc+Wild [27 Classes]	Resized to 224, Rotation, Flip, Histogram Equalization	Conv-Next-Base	AdamW	20	73.73%
Plant_Doc+Wild [27 Classes]	Resized to 224, YOLO v11, Rotation, Flip, Hist. Eq	YOLO26-L	AdamW	100	72.5%
Plant_Doc+Wild [27 Classes]	Resized to 224, YOLO v11, Rotation, Flip, Hist. Eq	YOLO26-X	AdamW	100	70.3%

Optimizer Note

- **AdamW (lr = 0.0001)** was used as the standard.
- Models with 40 epochs used a learning rate scheduler (0.0001 for 20 epochs, then 0.00001/0.00005 for the final 20).

5. Conclusions

CNN-Based Classifiers (e.g. Conv-Next, EfficientNet-B4) performed better in comparison to Transformer-Based (e.g. ViT-Vision Transformers) classifiers.

The classifiers for **Plant_Doc + Plant_Wild [Potato+Tomato] dataset** performed better when the original images were cropped to focus on their leaves rather than passing them straight through the classifier.

Rice Disease Dataset contained much clearer as well as a large number of samples which helped the classifier detect the features properly and hence the higher accuracy.

Grad-CAM images for 5 correctly classified examples per class for both **Plant_Doc+Plant_Wild dataset** as well as **Rice Disease dataset** have been included in the **'compilation_of_models_testing_dataset.ipynb'** (present in the repository).
